

(1)

① Two State Problem

Let us write the states as $|1\rangle \equiv \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ $|2\rangle \equiv \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

Then

$$H_0 = \begin{pmatrix} E_1 & 0 \\ 0 & E_2 \end{pmatrix} = \frac{E_1 + E_2}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \frac{E_1 - E_2}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Similarly

$$H_1 = H_0 + \gamma \begin{pmatrix} 0 & e^{i\omega t} \\ e^{-i\omega t} & 0 \end{pmatrix}$$

$$H = \frac{E_1 + E_2}{2} \mathbb{1} + \gamma \cos \omega t \sigma_1 - \gamma \sin \omega t \sigma_2 + \frac{E_1 - E_2}{2} \sigma_3$$

$$H = e^{\frac{i\omega t}{2} \sigma_3} \left[\frac{E_1 + E_2}{2} + \gamma \sigma_1 + \frac{E_1 - E_2}{2} \sigma_3 \right] e^{-\frac{i\omega t}{2} \sigma_3}$$

We used the "trick" discussed in the class. If

$|\psi(t)\rangle$ is the state that satisfies the equation

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle$$

We have shown $|\psi'(t)\rangle = e^{-\frac{i\omega t}{2} \sigma_3} |\psi(t)\rangle$ satisfies

the equation

$$i\hbar \frac{d}{dt} |\psi'(t)\rangle = H' |\psi'(t)\rangle$$

where

$$H' = \frac{E_1 + E_2}{2} + \gamma \sigma_1 + \frac{E_1 - E_2}{2} \sigma_3 + \frac{\hbar \omega}{2} \sigma_3$$

$$H_{\Lambda} = \frac{1}{2m} \left(\vec{P} - \frac{q}{c} \vec{A}_{\Lambda} \right)^2 + q\phi_{\Lambda}$$

$$= e^{-\frac{iq}{\hbar c} \Lambda} \left[\frac{1}{2m} \left(\vec{P} - \frac{q}{c} \vec{A} \right)^2 + q\phi + \frac{q}{c} \frac{\partial \Lambda}{\partial t} \right] e^{\frac{iq}{\hbar c} \Lambda}$$

$$= e^{-\frac{iq}{\hbar c} \Lambda} \left[H + \frac{q}{c} \frac{\partial \Lambda}{\partial t} \right] e^{\frac{iq}{\hbar c} \Lambda}$$

(iii) If $i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle$

and $i\hbar \frac{d}{dt} |\psi(t)\rangle_{\Lambda} = \hat{H}_{\Lambda} |\psi(t)\rangle_{\Lambda}$

$$\Rightarrow |\psi(t)\rangle_{\Lambda} = e^{-\frac{iq}{\hbar c} \Lambda} |\psi(t)\rangle$$

Substituting we see

$$\begin{aligned} i\hbar \frac{d}{dt} |\psi(t)\rangle_{\Lambda} &= \frac{q}{c} \frac{\partial \Lambda}{\partial t} |\psi(t)\rangle_{\Lambda} + e^{-\frac{iq}{\hbar c} \Lambda} \hat{H} e^{\frac{iq}{\hbar c} \Lambda} |\psi\rangle \\ &= \hat{H}_{\Lambda} |\psi(t)\rangle_{\Lambda} \end{aligned}$$

(2)

Let $E_1 - E_2 = \omega_{12} \hbar$, $\gamma = \omega_Y \hbar$, $\frac{E_1 + E_2}{2} = \omega_0 \hbar$

Then we see that

$$H' = \hbar \left(\omega_0 + \omega_Y \sigma_1 + \frac{1}{2}(\omega_{12} + \omega) \sigma_3 \right)$$

Let us define $\Omega = \left[\omega_Y^2 + (\omega_{12} + \omega)^2 / 4 \right]^{1/2}$

$$\eta_1 = \frac{\omega_Y}{\Omega} \quad \eta_2 = 0 \quad \eta_3 = \frac{1}{2} \frac{(\omega_{12} + \omega)}{\Omega}$$

Then

$$H' = \hbar \left(\omega_0 + \Omega \hat{n} \cdot \vec{\sigma} \right)$$

Now it is easy to solve for $| \psi'(t) \rangle$, we have

$$| \psi'(t) \rangle = \exp \left(\frac{-i}{\hbar} H' t \right) | \psi'(0) \rangle$$

$$e^{-i\omega_0 t} \left[\cos \Omega t - i \hat{n} \cdot \vec{\sigma} \sin \Omega t \right]$$

Initial state $| \psi'(0) \rangle = | \psi(0) \rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

Final state at time t $| \psi(t) \rangle = \exp \left(\frac{i\omega t}{2} \sigma_3 \right) | \psi'(t) \rangle$

$$| \psi(t) \rangle = e^{\frac{i\omega t}{2} \sigma_3} e^{-i\omega_0 t} \left[\cos \Omega t - i \hat{n} \cdot \vec{\sigma} \sin \Omega t \right] | \psi(0) \rangle$$

Amplitude to find the state in $| 2 \rangle$ is $\langle 2 | \psi(t) \rangle$ so

$$\begin{aligned} A = \langle 2 | \psi(t) \rangle &= (0 \ 1) e^{\frac{i\omega t}{2} \sigma_3} e^{-i\omega_0 t} \begin{pmatrix} \cos \Omega t \\ -i \hat{n} \cdot \vec{\sigma} \sin \Omega t \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ &= e^{\frac{-i\omega t}{2}} e^{-i\omega_0 t} (-i\eta_1 + \eta_2) \sin \Omega t \end{aligned}$$

Thus the probability

$$P_{1 \rightarrow 2}(t) = |A|^2 = \eta_1^2 \sin^2 \Omega t$$

$$= \frac{\gamma^2 / \hbar^2}{\gamma^2 / \hbar^2 + (\omega + \omega_{12})^2 / 4} \sin^2 \left\{ \left[\frac{\gamma^2}{\hbar^2} + \frac{(\omega + \omega_{12})^2}{4} \right]^{1/2} t \right\}$$

Problem 14.4.4

(3)

The Hamiltonian of an electron in a magnetic field

$$H = -\vec{\mu} \cdot \vec{B} \quad \vec{\mu} = -\frac{e}{2mc} \cdot 2 \vec{S}$$

g-factor of an electron.

$$H = -\frac{e\hbar}{mc} \vec{\sigma} \cdot \vec{B}$$

Now $\vec{B} = B \hat{i}$, which means $H = -\frac{e\hbar}{mc} B \sigma_1$.

Thus $|\psi(t)\rangle = \exp\left[-\frac{i}{\hbar} \frac{e\hbar B}{mc} t \sigma_1\right] |\psi(0)\rangle$

Let $\omega = \frac{eB}{mc}$, then the state at time t

$$|\psi(t)\rangle = (\cos \omega t - i \sigma_1 \sin \omega t) |\psi(0)\rangle$$

We are given $|\psi(0)\rangle \equiv \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, we want $|\psi(t)\rangle \equiv \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{i\alpha}$

We see that $\omega t = \pi \Rightarrow t = \frac{\pi}{\omega}$.

$$t = \frac{\pi mc}{eB}$$

we know $\frac{e\hbar}{2mc} \approx 0.6 \times 10^{-8} \text{ eV/G}$
 $B \approx 100 \text{ G}$

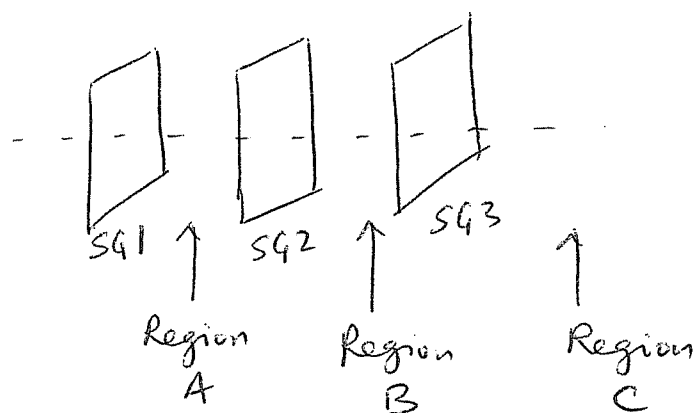
$$\hbar = 6.6 \times 10^{-16} \text{ eV} \cdot \text{s}$$

$$t = \frac{\pi \cdot 6.6 \times 10^{-16}}{2 \times 0.6 \times 10^{-8} \cdot 100}$$

$$t = \frac{\pi \hbar}{1.2 \times 10^{-6}} \text{ sec}$$

$$\approx 1.73 \times 10^{-9} \text{ sec}$$

Problem 14.5.3



Region A has particles ^{in state} $|\hat{z}, +\rangle$.

$$|\hat{z}, +\rangle = \frac{1}{\sqrt{2}} [|\hat{x}, +\rangle + |\hat{x}, -\rangle]$$

(i) If the second filter allows only $|\hat{x}, +\rangle$ states to pass. Then 50% of particles in region A will come out of region B.

(ii) If the 3rd filter allows only in $|\hat{z}, +\rangle$ then

Since $|\hat{x}, +\rangle = \frac{1}{\sqrt{2}} [|\hat{z}, +\rangle + |\hat{z}, -\rangle]$

only 50% of region B particles come into region C.
i.e... overall 25% of region A particles come to region C.

(iii) If the 2nd filter allows both spins up/down (no blocking) then Region B contains $|\hat{z}, +\rangle$ and then filter 3 transmits only $|\hat{z}, -\rangle$ then all particles will be blocked and 0% will reach region C.

Problem 15.1.2

(5)

If we take into account the electron and the proton spin, the Coulomb Hamiltonian ground state $E_0 = -R_y$ is four-fold degenerate!

(1) A Hyper fine interaction

$$H_{hf} = A \vec{S}_1 \cdot \vec{S}_2$$

$$= \frac{A}{2} [\vec{S}^2 - \vec{S}_1^2 - \vec{S}_2^2]$$

splits these levels. Since $\vec{S}_1$ and $\vec{S}_2$ are both spin- $\frac{1}{2}$, $\vec{S}$ can be spin-1 or spin-0.

$$E_{hf} = \frac{A}{2} \left[2\hbar^2 - \frac{3}{2}\hbar^2 \right] = \frac{\hbar^2 A}{4} \text{ triplet}$$
$$= \frac{A}{2} \left[0 - \frac{3}{2}\hbar^2 \right] = -\frac{3\hbar^2 A}{4} \text{ singlet.}$$

$$\therefore E_+ = -R_y + \frac{\hbar^2 A}{4} \text{ triplet}$$

$$E_- = -R_y - \frac{3\hbar^2 A}{4} \text{ singlet.}$$

(2) To estimate the frequency emitted we equate

$$h\nu = \Delta E = A\hbar^2$$

In order to express it as wave length we use

$$\frac{hc}{\lambda} = \Delta E$$

$$\Rightarrow \lambda = \frac{hc}{\Delta E}$$

(6)

Now $\Delta E = A \hbar^2 = \frac{1}{2} \frac{e^2 \hbar^2}{a_0^3} \frac{5.6}{M m c^2}$

$a_0 = \frac{\hbar^2}{m e^2}$ and $\frac{m e^4}{2 \hbar^2} \approx 13.3 \text{ eV}$

$$\begin{aligned} \Delta E &= \frac{5.6}{2} \frac{m^3 e^6}{\hbar^4} \frac{e^2 \hbar^2}{M m c^2} \\ &= \frac{5.6}{2} \left(\frac{m e^4}{2 \hbar^2} \right)^2 \cdot \frac{1}{M c^2} \\ &\approx 2.5.6 \cdot \frac{(13.3)^2 \text{ eV}^2}{1 \text{ GeV}} = 1.98 \times 10^{-6} \text{ eV} . \end{aligned}$$

$\lambda = \frac{h c}{\Delta E} \approx 62 \text{ cm} .$

(3) The probability ratio $P(\text{triplet})/P(\text{singlet}) = \overset{\text{degeneracy}}{3} \frac{\exp(-E_{\text{triplet}}/kT)}{\exp(-E_{\text{singlet}}/kT)}$

$\approx 3 e^{-\Delta E/kT}$

$kT \approx \frac{1}{40} \text{ eV}$ at $T = 300 \text{ K}$ (room temperature) .

Since $\Delta E \ll kT$ we see that

$P(\text{triplet})/P(\text{singlet}) \approx 3 .$

Exercise 15.2.2

$$(1) \quad j_1 = 1 \quad j_2 = \frac{1}{2} \Rightarrow j = \frac{3}{2} \text{ or } \frac{1}{2}$$

$$\begin{array}{ccc} & \rightarrow j & \\ \downarrow m & & \\ & j = \frac{3}{2} & j = \frac{1}{2} \\ & | \frac{3}{2} \frac{3}{2} \rangle & \\ & | \frac{3}{2} \frac{1}{2} \rangle & | \frac{1}{2} \frac{1}{2} \rangle \\ & | \frac{3}{2} -\frac{1}{2} \rangle & | \frac{1}{2} -\frac{1}{2} \rangle \\ & | \frac{3}{2} -\frac{3}{2} \rangle & \end{array}$$

$$\rightarrow | \frac{3}{2} \frac{3}{2} \rangle = | 1 1 \rangle \otimes | \frac{1}{2} \frac{1}{2} \rangle$$

Applying J_- on both sides we get-

$$\sqrt{3} \hbar | \frac{3}{2} \frac{1}{2} \rangle = \sqrt{2} \hbar | 1 0 \rangle \otimes | \frac{1}{2} \frac{1}{2} \rangle + \hbar | 1 1 \rangle \otimes | \frac{1}{2} -\frac{1}{2} \rangle$$

$$\rightarrow | \frac{3}{2} \frac{1}{2} \rangle = \sqrt{\frac{2}{3}} | 1 0, \frac{1}{2} \frac{1}{2} \rangle + \frac{1}{\sqrt{3}} | 1 1, \frac{1}{2} -\frac{1}{2} \rangle$$

$$\rightarrow | \frac{3}{2}, -\frac{1}{2} \rangle = \sqrt{\frac{2}{3}} | 1 0, \frac{1}{2} -\frac{1}{2} \rangle + \frac{1}{\sqrt{3}} | 1 -1, \frac{1}{2} \frac{1}{2} \rangle$$

$$\rightarrow | \frac{3}{2}, -\frac{3}{2} \rangle = | 1 -1, \frac{1}{2} -\frac{1}{2} \rangle$$

$$\rightarrow | \frac{1}{2} \frac{1}{2} \rangle = \sqrt{\frac{2}{3}} | 1 1, \frac{1}{2} -\frac{1}{2} \rangle - \sqrt{\frac{1}{3}} | 1 0, \frac{1}{2} \frac{1}{2} \rangle$$

$$\rightarrow | \frac{1}{2} -\frac{1}{2} \rangle = \sqrt{\frac{1}{3}} | 1 0, \frac{1}{2} -\frac{1}{2} \rangle - \sqrt{\frac{2}{3}} | 1 -1, \frac{1}{2} \frac{1}{2} \rangle$$

One can read off the Clebsch-Gordan coefficients.

$$(2) \quad j_1 = 1 \quad j_2 = 1 \Rightarrow j = 2, 1, 0$$

$$\begin{array}{ccc}
 \downarrow m & \xrightarrow{j} & \\
 & j=2 & j=1 \quad j=0 \\
 & |2, 2\rangle & \\
 & |2, 1\rangle & |1, 1\rangle \\
 & |2, 0\rangle & |1, 0\rangle \quad |0, 0\rangle \\
 & |2, -1\rangle & |1, -1\rangle \\
 & |2, -2\rangle &
 \end{array}$$

$$\rightarrow |2, 2\rangle = |1, 1, 1\rangle$$

Applying J_- on both sides.

$$2\hbar |2, 1\rangle = \hbar\sqrt{2} |1, 0, 1\rangle + \hbar\sqrt{2} |1, 1, 0\rangle$$

$$\rightarrow |2, 1\rangle = \frac{1}{\sqrt{2}} |1, 0, 1\rangle + \frac{1}{\sqrt{2}} |1, 1, 0\rangle$$

$$\begin{aligned}
 \sqrt{3} \cdot \sqrt{2} \hbar |2, 0\rangle &= \frac{\hbar\sqrt{2}}{\sqrt{2}} |1, -1, 1\rangle + \frac{\hbar\sqrt{2}}{\sqrt{2}} |1, 0, 1\rangle \\
 &\quad + \frac{\hbar\sqrt{2}}{\sqrt{2}} |1, 0, 0\rangle + \frac{\hbar\sqrt{2}}{\sqrt{2}} |1, 1, -1\rangle
 \end{aligned}$$

$$\rightarrow |2, 0\rangle = \frac{1}{\sqrt{6}} |1, -1, 1\rangle + \frac{\sqrt{2}}{3} |1, 0, 1\rangle + \frac{1}{\sqrt{6}} |1, 1, -1\rangle$$

$$\rightarrow |2, -1\rangle = \frac{1}{\sqrt{2}} |1, 0, -1\rangle + \frac{1}{\sqrt{2}} |1, -1, 0\rangle$$

$$\rightarrow |2, -2\rangle = |1, -1, -1\rangle$$

$$\rightarrow |1, 1\rangle = \frac{1}{\sqrt{2}} |1, 1, 0\rangle - \frac{1}{\sqrt{2}} |1, 0, 1\rangle$$

$$\begin{aligned}
 \hbar\sqrt{2} |1, 0\rangle &= \frac{\hbar\sqrt{2}}{\sqrt{2}} |1, 0, 0\rangle + \frac{\hbar\sqrt{2}}{\sqrt{2}} |1, 1, -1\rangle \\
 &\quad - \frac{\hbar\sqrt{2}}{\sqrt{2}} |1, -1, 1\rangle - \frac{\hbar\sqrt{2}}{\sqrt{2}} |1, 0, 1\rangle
 \end{aligned}$$

$$\rightarrow |1, 0\rangle = \frac{1}{\sqrt{2}} |1, 1, -1\rangle - \frac{1}{\sqrt{2}} |1, -1, 1\rangle$$

$$\rightarrow |1-1\rangle = \frac{1}{\sqrt{2}} |10, 1-1\rangle - \frac{1}{\sqrt{2}} |1-1, 10\rangle$$

(9)

We next want $|0,0\rangle$ which has the form

$$|00\rangle = \alpha |11, 1-1\rangle + \beta |10, 10\rangle + \gamma |1-1, 11\rangle$$

We want $|00\rangle$ to be orthogonal to $|10\rangle$ and $|20\rangle$.

$$\begin{aligned} \langle 10|00\rangle = 0 &\Rightarrow \alpha = \gamma \\ \langle 20|00\rangle = 0 &\Rightarrow \alpha + \gamma + 2\beta = 0 \end{aligned}$$

which means $\alpha = \gamma = -\beta = \frac{1}{\sqrt{3}}$

$$|00\rangle = \frac{1}{\sqrt{3}} |11, 1-1\rangle - \frac{1}{\sqrt{3}} |10, 10\rangle + \frac{1}{\sqrt{3}} |1-1, 11\rangle$$

15.2.5

(1) First notice that $\vec{S}_1 = \vec{\sigma}_1 \frac{\hbar}{2}$ and $\vec{S}_2 = \vec{\sigma}_2 \frac{\hbar}{2}$.

$$P_1 = \frac{3}{4} I + \frac{\vec{S}_1 \cdot \vec{S}_2}{\hbar^2} = \left(\frac{3}{4} I + \frac{1}{4} \vec{\sigma}_1 \cdot \vec{\sigma}_2 \right)$$

$$P_1 P_1 = \left(\frac{9}{16} I + \frac{6}{16} \vec{\sigma}_1 \cdot \vec{\sigma}_2 + \frac{1}{16} (\vec{\sigma}_1 \cdot \vec{\sigma}_2)^2 \right)$$

$$\begin{aligned} \text{Now } (\vec{\sigma}_1 \cdot \vec{\sigma}_2) (\vec{\sigma}_1 \cdot \vec{\sigma}_2) &= \underbrace{(\vec{\sigma}_1 \cdot \vec{\sigma}_1)}_3 + i \underbrace{(\vec{\sigma}_1 \times \vec{\sigma}_1)}_{2i\vec{\sigma}_1} \cdot \vec{\sigma}_2 \\ &= (3I - 2\vec{\sigma}_1 \cdot \vec{\sigma}_2) \end{aligned}$$

$$\therefore P_1^2 = \left(\frac{3}{4} I + \frac{6}{16} \vec{\sigma}_1 \cdot \vec{\sigma}_2 - \frac{2}{16} \vec{\sigma}_1 \cdot \vec{\sigma}_2 \right) = P_1$$

$$P_2 = \left(\frac{1}{4} I - \frac{1}{4} \vec{\sigma}_1 \cdot \vec{\sigma}_2 \right)$$

$$P_2 P_2 = \left(\frac{1}{16} I - \frac{2}{16} \vec{\sigma}_1 \cdot \vec{\sigma}_2 + \frac{1}{16} (3I - 2\vec{\sigma}_1 \cdot \vec{\sigma}_2) \right) = P_2$$

(10)

$$\begin{aligned}
 P_1 P_2 &= \left(\frac{3}{4} I + \frac{1}{4} \vec{\sigma}_1 \cdot \vec{\sigma}_2 \right) \left(\frac{1}{4} I - \frac{1}{4} \vec{\sigma}_1 \cdot \vec{\sigma}_2 \right) \\
 &= \left(\frac{3}{16} I - \frac{2}{16} (\vec{\sigma}_1 \cdot \vec{\sigma}_2) - \frac{1}{16} (3I - 2\vec{\sigma}_1 \cdot \vec{\sigma}_2) \right) \\
 &= 0
 \end{aligned}$$

(2) Now let us write $\vec{S} = \vec{S}_1 + \vec{S}_2$; $\vec{S}_1^2 = \frac{3}{4} I = \vec{S}_2^2$

$$\therefore \frac{(\vec{S}_1 \cdot \vec{S}_2)}{\hbar^2} = \frac{1}{2\hbar^2} (\vec{S}^2 - \vec{S}_1^2 - \vec{S}_2^2) = \frac{\vec{S}^2}{2\hbar^2} - \frac{3}{4} I$$

$$\therefore P_1 = \frac{\vec{S}^2}{2\hbar^2}$$

Now if P_1 acts on $|11\rangle$, $|10\rangle$ or $|1-1\rangle$ we get 1.

If it acts on $|00\rangle$ we get 0.

$$P_2 = I - \frac{\vec{S}^2}{2\hbar^2}$$

If P_2 acts on $|11\rangle$, $|10\rangle$, $|1-1\rangle$ we get a 0
and when it acts on $|00\rangle$ we get 1.

Thus P_1 projects into the spin-1 space and P_2 projects into spin-0.